

THE ℓ_1^d OBSTRUCTION TO THE LITERAL ARBITRARY-NORM STEINITZ $O(\sqrt{d})$ CLAIM

ARGUS OPEN RESEARCH PROJECT

ABSTRACT. AIM Problem 1.4 states an $O(\sqrt{d})$ bound for the Steinitz rearrangement constant while explicitly quantifying over an arbitrary norm. We give a self-contained verification of the classical Grinberg–Sevast’yanov obstruction in ℓ_1^d . For every $d \geq 1$ and every sufficiently large odd integer K , an explicit finite indexed zero-sum family in the ℓ_1^d unit ball forces every ordering to have a partial sum of norm at least

$$\frac{d+1}{2} - \frac{d-1}{2K}.$$

Consequently $S(B_1^d) \geq (d+1)/2$, which contradicts a dimension-uniform $O(\sqrt{d})$ bound over all norms. The argument concerns the ordinary zero-sum vector-permutation constant and uses no matrix or sign-sequence transfer. It does not refute the genuinely Euclidean conjecture $S(B_2^d) = O(\sqrt{d})$.

1. THE STATEMENT AND ITS SCOPE

For a centrally symmetric convex body $B \subset \mathbb{R}^d$ with 0 in its interior, write $\|\cdot\|_B$ for its Minkowski norm. The Steinitz constant $S(B)$ is the least C such that every finite indexed family $V = (v_1, \dots, v_n) \subset B$ with $\sum_i v_i = 0$ admits a permutation π satisfying

$$\left\| \sum_{i=1}^k v_{\pi(i)} \right\|_B \leq C \quad (1 \leq k \leq n).$$

Equivalently, $S(B)$ is the supremum, over finite zero-sum families in B , of the least possible maximum partial-sum norm.

The displayed formula in AIM Problem 1.4 quantifies over “a norm X ” and asserts $S(X, d) = O(\sqrt{d})$ [1]. The official discussion separately treats arbitrary norms and the Euclidean norm. Modern terminology makes the distinction explicit: Ambrus and Heck define the general constant $S(B)$, recall $S(B) \leq d$, record the lower bound $S(B_1^d) \geq (d+1)/2$, and state the Euclidean assertion $S(B_2^d) = O(\sqrt{d})$ as a separate conjecture [2].

Theorem 1.1. *For every integer $d \geq 1$,*

$$S(B_1^d) \geq \frac{d+1}{2}.$$

Consequently no bound $S(B) = O(\sqrt{d})$, with an absolute implied constant, can hold uniformly over all norms on $\mathbb{R}^d$.

Remark 1.2 (Boundary of the conclusion). The theorem refutes the literal arbitrary-norm assertion. It does not refute the Euclidean conjecture

$$S(B_2^d) = O(\sqrt{d}).$$

The word “Euclidean” in a secondary catalogue title cannot narrow the explicit arbitrary-norm quantifier in the source formula.

2. THE GRINBERG–SEVAST’YANOV FAMILY

Fix $d \geq 1$ and put $r = d - 1$. Let K be an odd integer satisfying $K \geq \max\{1, r/2\}$, and define

$$\tau_K = 1 - \frac{r}{2K} \geq 0.$$

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Let $e_1, \dots, e_r$ be the first r coordinate vectors in $\mathbb{R}^d$. For $s = 1, \dots, K$, introduce separately indexed vectors

$$a_s = \left(-\frac{1}{2K}, \dots, -\frac{1}{2K}, \tau_K \right),$$

$$b_s = \left(-\frac{1}{2K}, \dots, -\frac{1}{2K}, -\tau_K \right),$$

where the first r coordinates are all $-1/(2K)$. Set

$$V_K = (e_1, \dots, e_r, a_1, \dots, a_K, b_1, \dots, b_K).$$

This is the lower-bound family printed by Grinberg and Sevast'yanov [3]. We verify every required condition directly.

Lemma 2.1. *Every vector in V_K has ℓ_1 norm 1, and the indexed family has total sum zero.*

Proof. Clearly $\|e_j\|_1 = 1$. Moreover,

$$\|a_s\|_1 = \|b_s\|_1 = \frac{r}{2K} + \tau_K = 1.$$

In each of the first r coordinates, the $2K$ dense vectors contribute $2K(-1/(2K)) = -1$, which is cancelled by the corresponding coordinate vector. In the final coordinate, the a_s and b_s contribute $K\tau_K - K\tau_K = 0$. $\square$

3. EVERY ORDERING HAS A LARGE PARTIAL SUM

Lemma 3.1. *For every permutation of V_K , some partial sum has ℓ_1 norm at least*

$$\frac{d+1}{2} - \frac{d-1}{2K}.$$

Proof. Fix an arbitrary permutation and stop immediately after exactly K of the $2K$ dense vectors a_s, b_s have appeared. Let A and B be the numbers of a - and b -vectors in this prefix. Then $A + B = K$.

In each of the first r coordinates, the dense vectors contribute $K(-1/(2K)) = -1/2$. If the corresponding e_j has appeared, that coordinate of the partial sum is $1/2$; otherwise it is $-1/2$. Thus the first r coordinates contribute exactly $r/2$ to the ℓ_1 norm, independently of the positions of the e_j .

The last coordinate is $(A - B)\tau_K$. Since $K = A + B$ is odd, $A - B$ is odd and hence nonzero. Therefore $|A - B| \geq 1$, and the selected partial sum has norm at least

$$\frac{r}{2} + \tau_K = \frac{d+1}{2} - \frac{d-1}{2K}.$$

$\square$

Proof of Theorem 1.1. Let $C < (d+1)/2$. Choose a sufficiently large odd K . By Lemma 2.1, V_K is a finite zero-sum family in the ℓ_1^d unit ball. By Lemma 3.1, every ordering has a partial sum of norm greater than C . Taking the supremum over finite families gives $S(B_1^d) \geq (d+1)/2$.

Finally,

$$\frac{S(B_1^d)}{\sqrt{d}} \geq \frac{d+1}{2\sqrt{d}} \longrightarrow \infty,$$

so no arbitrary-norm $O(\sqrt{d})$ estimate can hold. $\square$

Remark 3.2. The family is not the ineffective collection $\{e_i, -e_i\}$, which can be cancelled pairwise. For $d > 1$ one has

$$a_s + b_t = (-1/K, \dots, -1/K, 0),$$

so alternating dense vectors cancels only the final coordinate while accumulating drift in every other coordinate.

4. SOURCE ALIGNMENT

The opening paragraph of the 1980 paper defines the ordinary zero-sum permutation constant in an arbitrary d -dimensional normed space, and its displayed family is the one used above [3]. The modern account [2] independently records the same ℓ_1^d lower bound and separates it from the Euclidean conjecture. Thus no transfer from a matrix Steinitz lemma, a sign-selection problem, or a nonstandard ordering constant is involved.

The proof of Theorem 1.1 is fully symbolic and requires no numerical experiment.

APPENDIX A. CHECKLIST FOR AN INDEPENDENT REFEREE

- (1) Confirm that AIM Problem 1.4 quantifies over an arbitrary norm and concerns ordinary zero-sum vector permutations.
- (2) Verify the ℓ_1 unit-norm calculation for e_j, a_s, b_s .
- (3) Check the zero-sum identity coordinate by coordinate.
- (4) In an arbitrary ordering, stop after the K th dense vector and verify that every one of the first $d - 1$ coordinates has absolute value $1/2$.
- (5) Use the oddness of K to prove $|A - B| \geq 1$ in the last coordinate.
- (6) Let $K \rightarrow \infty$ and distinguish the arbitrary-norm conclusion from the Euclidean conjecture.

REFERENCES

- [1] American Institute of Mathematics, *Hereditary discrepancy and related topics*, Problem 1.4, AimPL problem-list export.
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